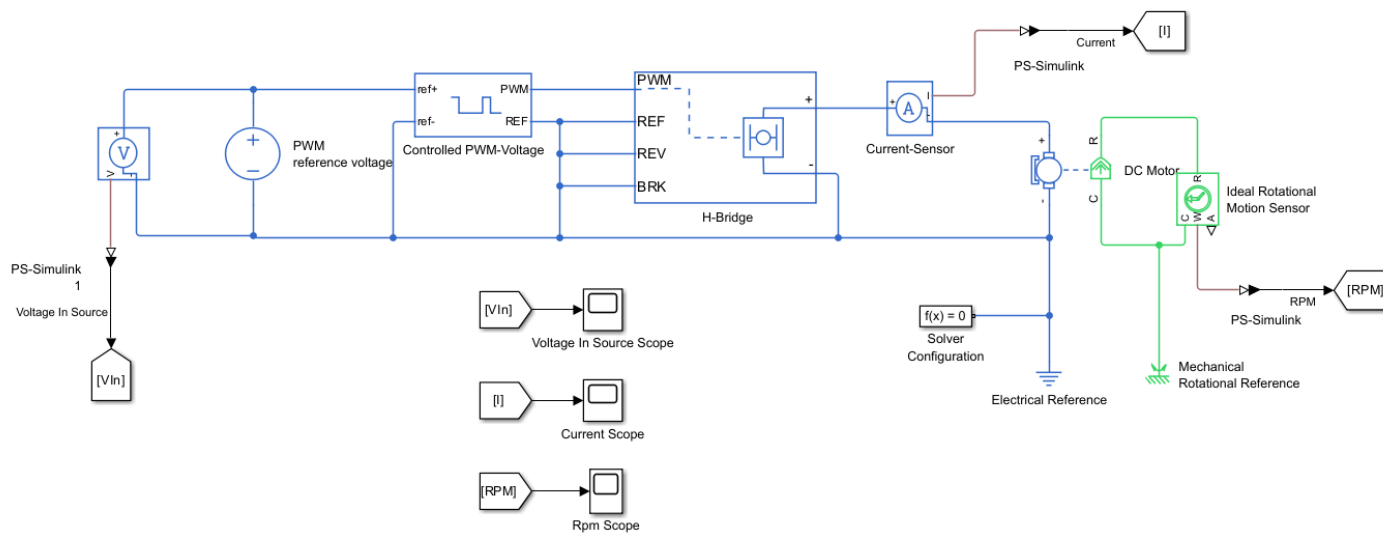


DC Motor Control with H-Bridge and PWM

System Overview (Main Diagram)



Component Explanations

1. PWM Reference Voltage (DC Voltage Source)

This block provides the **main control signal** for the system. It outputs a simple, constant DC voltage that tells the PWM generator what duty cycle (on-off ratio) to create.

2. Controlled PWM Voltage

This block is the **"brain" of the PWM generation**. It reads the input voltage (from the "PWM Reference Voltage" block) and creates a high-frequency, pulsed voltage signal (a PWM waveform) based on it.

3. H-Bridge

This block is the **"muscle" or power switch**. It takes the low-power PWM signal and uses it to rapidly switch the main motor power on and off. It's what allows the PWM signal to control the high-power motor, and it's also what makes it possible to reverse the motor's direction (though not used in this project).

4. DC Motor

This is the main component of the project. It's a **physical model of a motor** that converts the electrical energy from the H-Bridge into mechanical rotational motion (spinning).

5. Current Sensor

This is a **measurement device**. It's placed in line with the motor to measure the amount of electrical current flowing into it.

6. Ideal Rotational Motion Sensor

This is another **measurement device**. It attaches to the motor's spinning shaft to measure its rotational speed (RPM).

7. PS-Simulink Converter

This is a **"translator" block**. Simscape (physical) signals and regular Simulink signals are different. This block converts the physical measurements (like Amps or RPM) into standard Simulink signals that the "Scope" blocks can understand and display.

8. Scope (Current, RPM)

This is a **visual plotter**. It takes the signal from the PS-Simulink Converter and draws a graph of its value over time, allowing you to see the results.

9. Electrical Reference

This is the **electrical "ground"**. It defines the zero-volt (0V) reference point for the entire electrical circuit. Every electrical component needs this to know what its voltage is measured relative to.

10. Mechanical Rotational Reference

This is the **mechanical "ground"**. It defines a stationary, non-moving reference point. The motor's casing is connected to this, so the motor's shaft spins *relative* to this stationary frame.

11. Solver Configuration

This is the **simulation "engine"**. It's a required block for all Simscape models. It sets the mathematical rules and precision (like tolerances and step size) that Simulink uses to calculate the physical interactions in your model.

Results

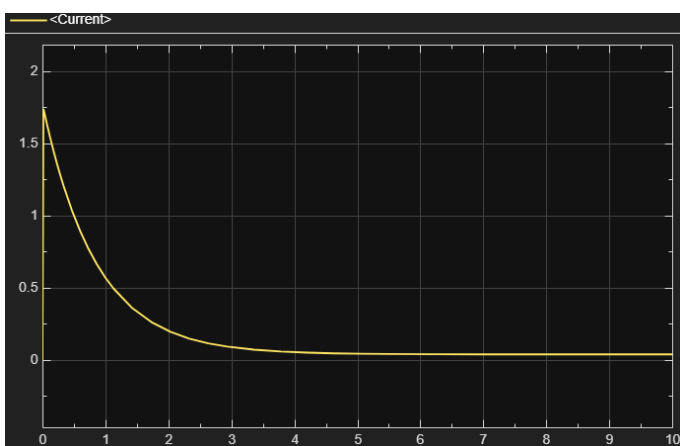


Figure 1: Current Graph (A)

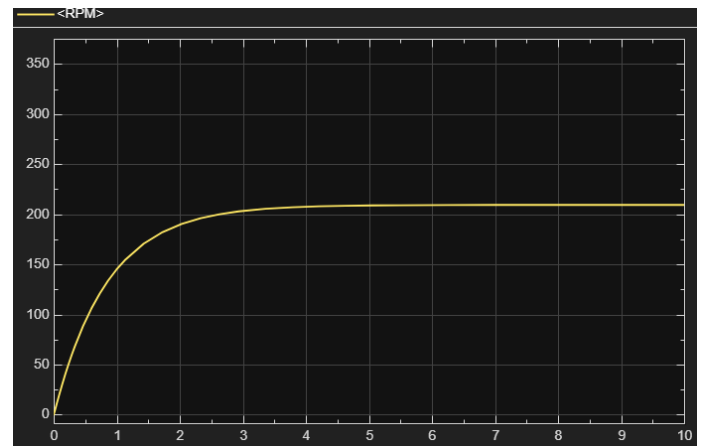


Figure 2: RPM Graph